Skia canvas / typography lab

One scene: raster, recorded picture, PDF and SVG. Text exported as paths.

01 / Pair kerning

Times New Roman / separate characters vs shaped run

SEPARATE CHARACTERS

AVATAR To WA

SHAPED RUN / PAIR ADJUSTMENTS

AVATAR To WA

Widths: 342,4 -> 311,8 px / 12 glyphs

02 / Context and ligatures

Arial / Arabic joining and lam-alef substitution

ISOLATED LETTERS

س ل ا م

JOINED WORD

سلام

LAM + ALEF

لا

Lam-alef: 2 UTF-16 units -> 1 glyph(s)

03 / Combining marks + scripts

Precomposed / decomposed accents; Cyrillic and Greek

Café / Café

Привет / Καλημέρα

Å ñ ä ô

UTF-16 pair 😀

e + combining acute: 2 units -> 1 glyph(s)

04 / Bidirectional layout

Logical input / visual glyph order / punctuation and digits

ABC 123 - שלום

XYZ / 2026 سلام

سلام / (שלום)

Contours retain joining and visual glyph order.

05 / Transform + clip + state

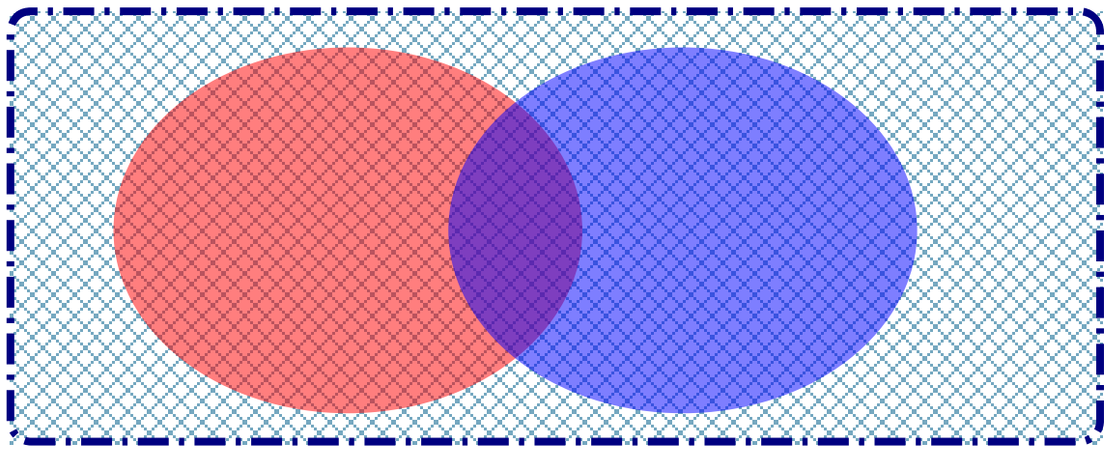
Rotated, scaled glyph outlines inside a fixed viewport



After RestoreState: origin = (0, 0)

06 / Alpha + patterns + dashes

Layered fills, GDI hatching and custom dash pattern



130/255 alpha; dash = [12, 4, 2, 4].

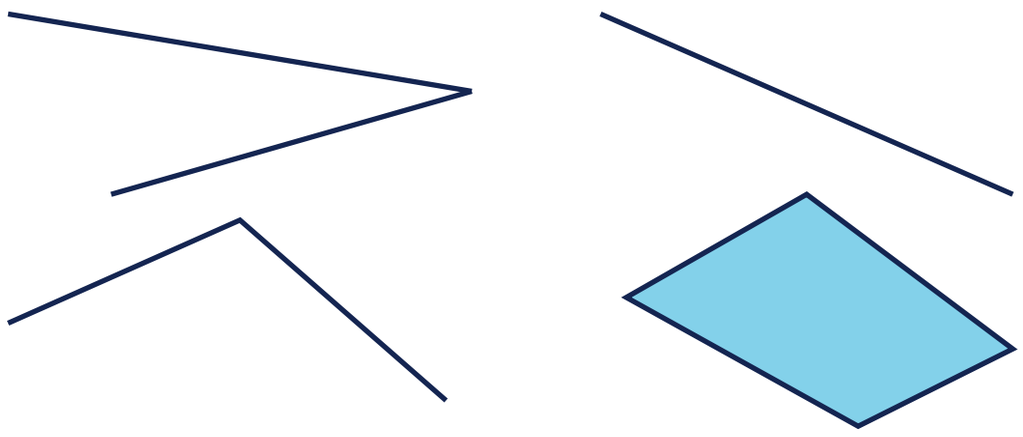
Native Windows: GDI outlines + Uniscribe shaping. No SVG font dependency.

Skia canvas / complete API atlas

02-primitives / native results and compatibility operations

Lines + polygonal paths

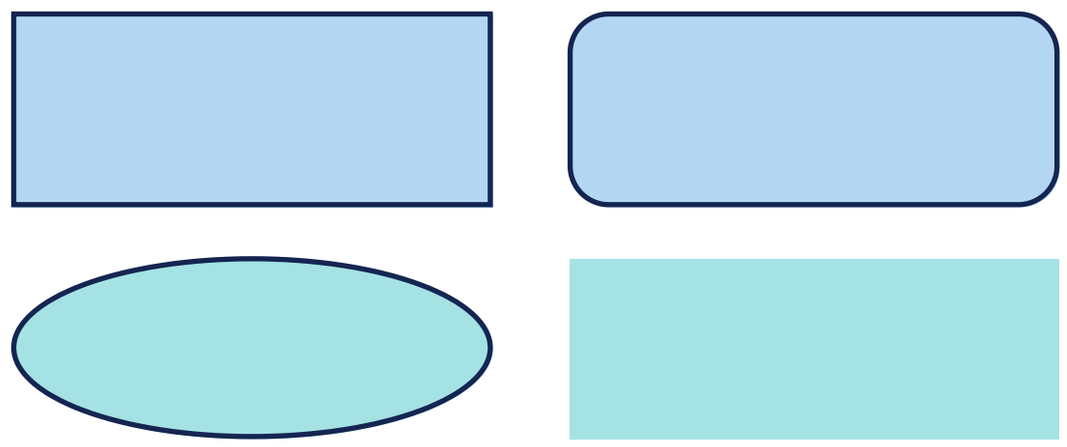
MoveTo / LineTo / PenPos / Polyline / Polygon



PenPos = (384, 408)

Closed shapes + fill / stroke

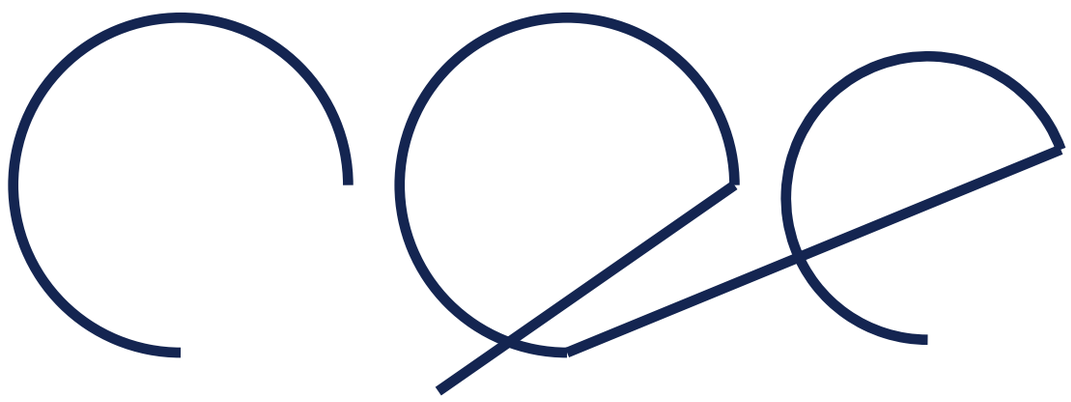
Rectangle / RoundRect / Ellipse / FillRect



Stroke + fill are independent; bsClear removes fill.

Arcs + current position

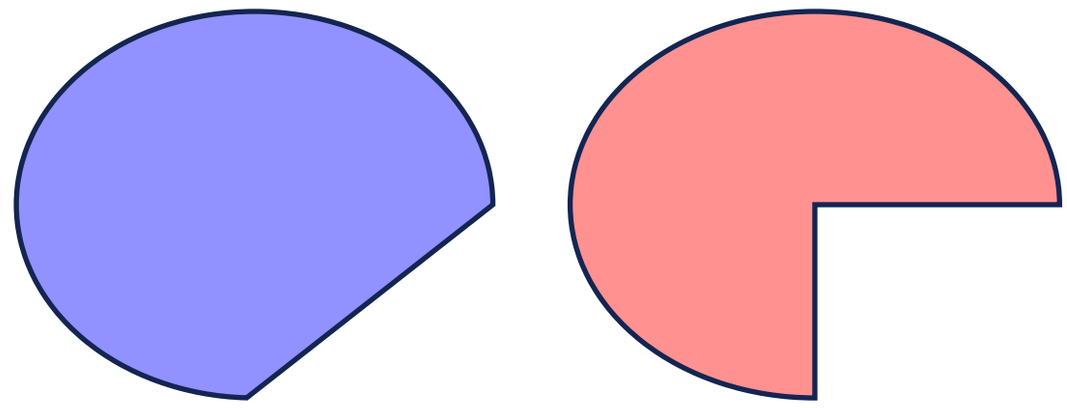
Arc / ArcTo / AngleArc (VCL angle convention)



ArcTo connects from PenPos; AngleArc uses degrees.

Arc closures

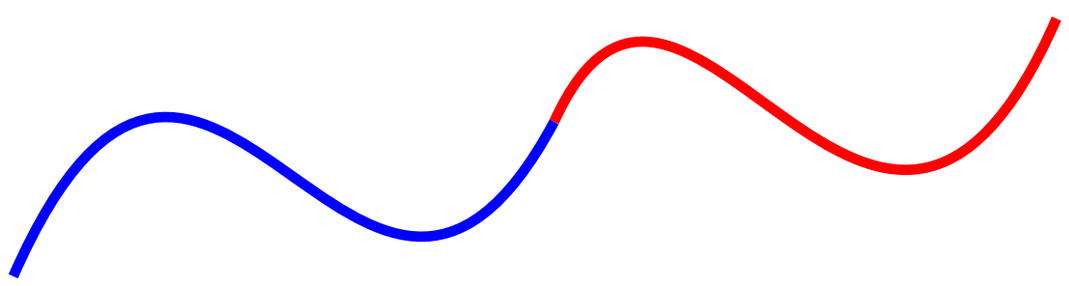
Chord / Pie / overlapping translucent interiors



Chord closes endpoints; Pie connects through center.

Cubic curves

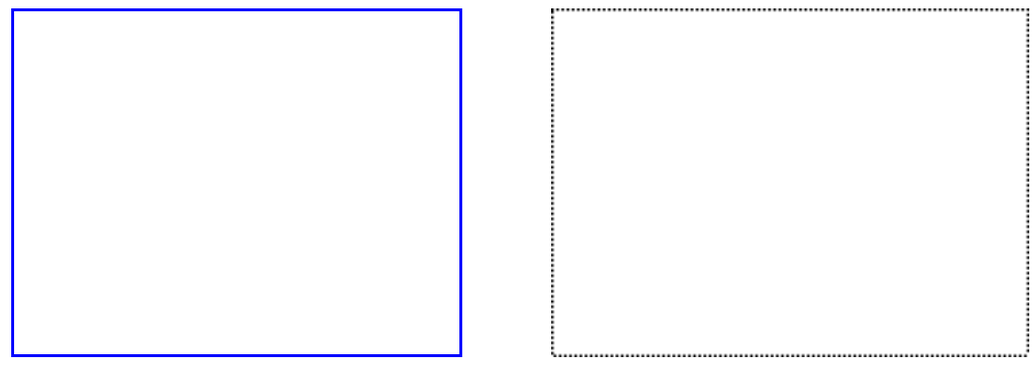
PolyBezier / PolyBezierTo



BezierTo begins at the current PenPos.

Frames + focus + refresh

FrameRect / DrawFocusRect / Refresh



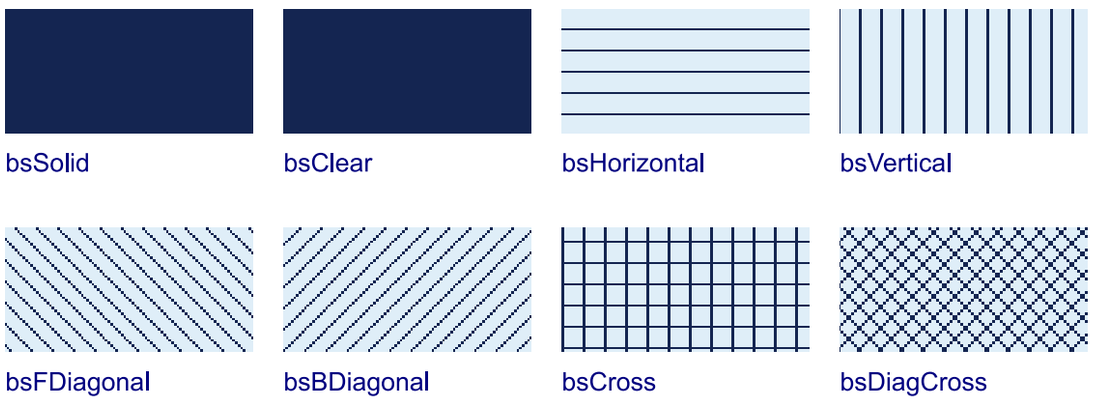
Refresh is a no-op: drawing is immediate.

Skia canvas / complete API atlas

03-brushes / native results and compatibility operations

All eight brush styles

TBrushStyle / identical background and foreground



bsClear leaves the destination unchanged.

Horizontal + vertical gradients

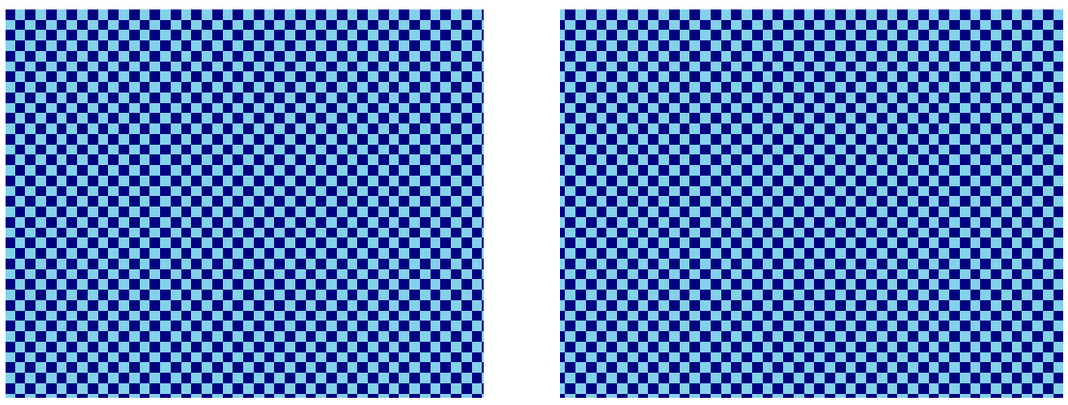
GradientStartColor / GradientEndColor



Gradient coordinates follow each shape bounds.

Bitmap patterns

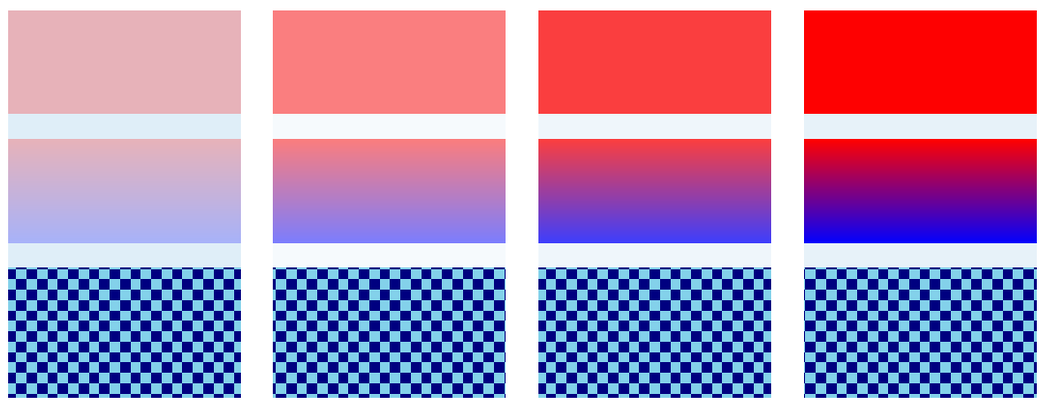
Pattern property / Assign(TBitmap) / ownership copy



Original bitmap can be freed after assigning the brush.

Alpha ramp

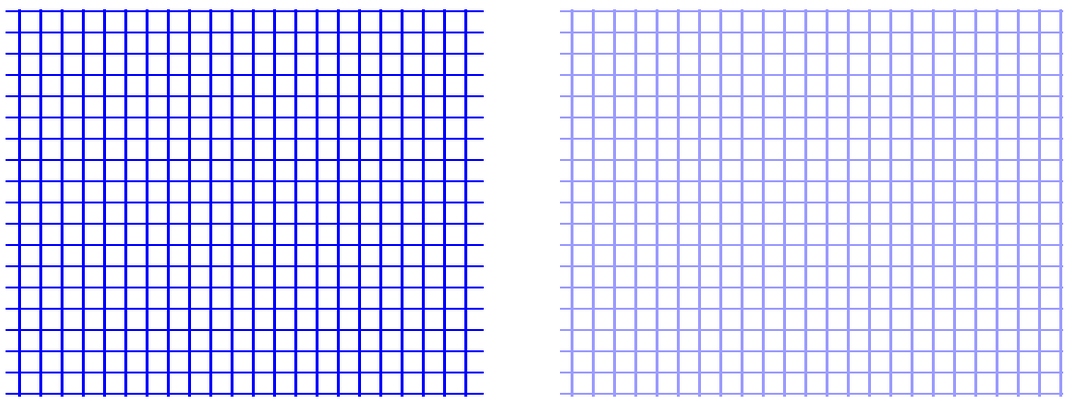
Solid / gradient / bitmap pattern opacity



Columns: alpha = 64, 127, 190, 253.

Assign + visibility

VCL TBrush / TSkiaBrush / IsVisible



Assigned brush is visible: True

Resource cache + invalidation

AcquireFillPaint / InvalidateBrush / InvalidateAll

Same key -> same paint

Color / alpha / bounds -> new paint

OnChange invalidates canvas resources

Cache identity is also checked in runtime diagnostics.

Skia canvas / complete API atlas

04-pens / native results and compatibility operations

All seven pen styles

Pen.Style / Width = 2

psSolid

psDash

psDot

psDashDot

psDashDotDot

psClear

psInsideFrame

psInsideFrame keeps the stroke inside closed shape bounds.

Width / alpha / custom dashes

Widths 1, 5, 9, 13; user dash [12,4,2,4]

Alpha, stroke width and user dash are independent.

Stroke caps

Butt / Round / Square; vertical guides show endpoints



Caps extend beyond / stop at the geometric endpoints.

Stroke joins + miter limit

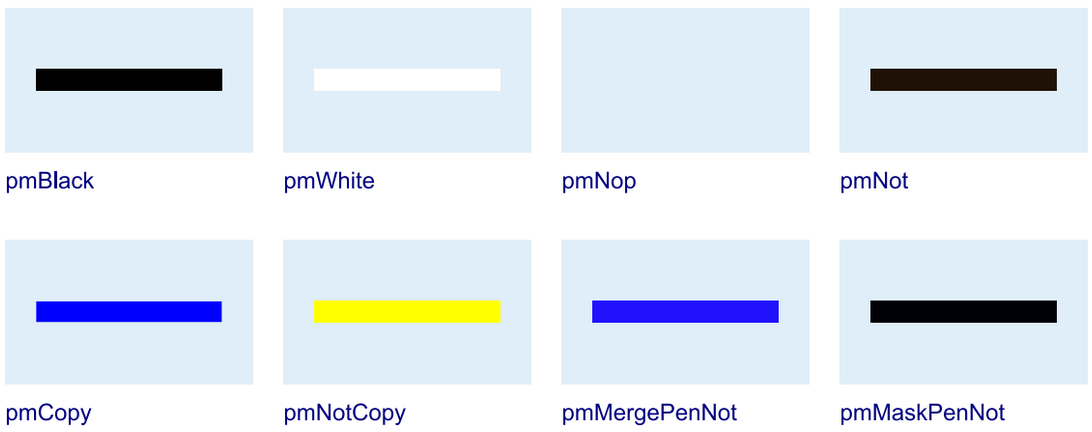
Miter / Round / Bevel / clipped miter



Pen.Assign(VCL) / IsVisible = True

Pen.Mode: 0..7

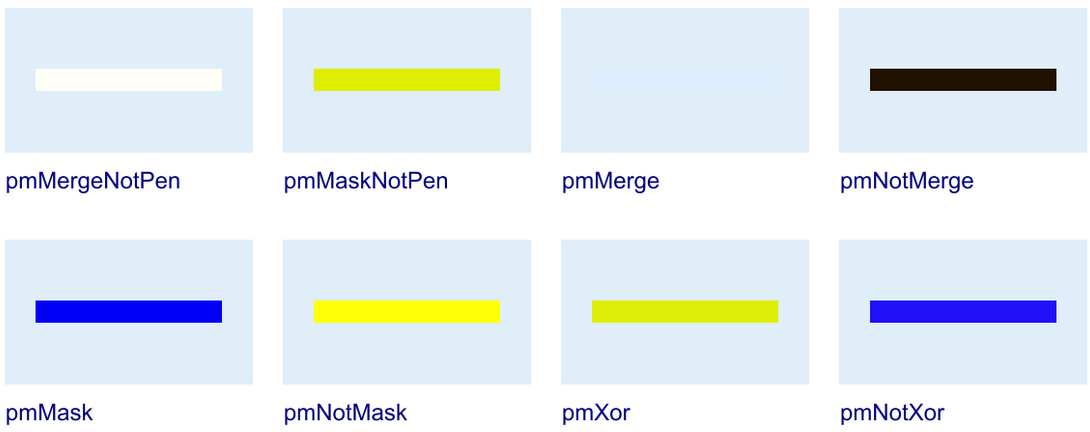
All 16 Boolean combinations of source and destination



Each mode changes the same blue stroke over the backdrop.

Pen.Mode: 8..15

All 16 Boolean combinations of source and destination



Each mode changes the same blue stroke over the backdrop.

Skia canvas / complete API atlas

05-transfers / native results and compatibility operations

Source / Draw / opacity

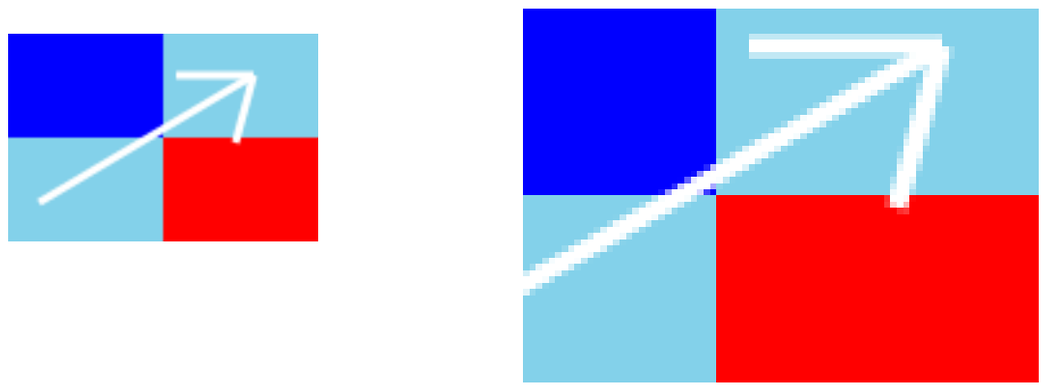
Asymmetric source makes cropping and flipping visible



Draw(TBitmap) / Draw(...,Opacity) / StretchDraw.

BitBlt + CopyRect

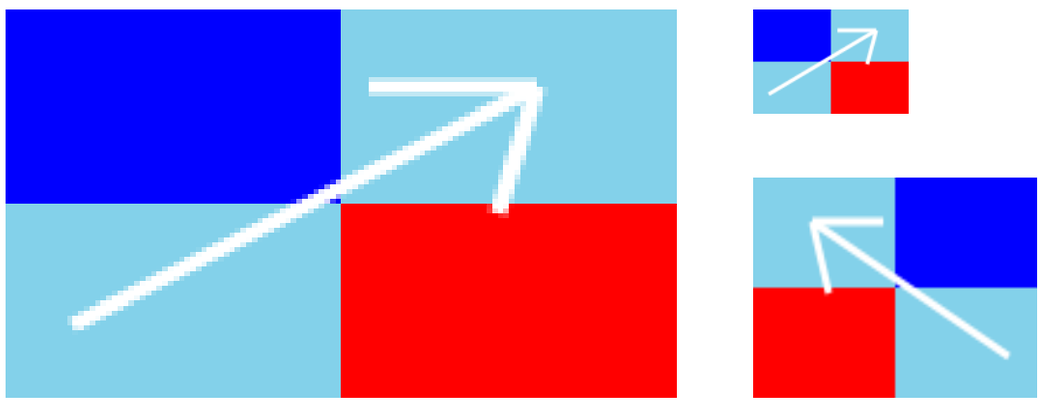
Full copy and cropped source rectangle



CopyMode controls the Boolean source/destination operation.

StretchBlt

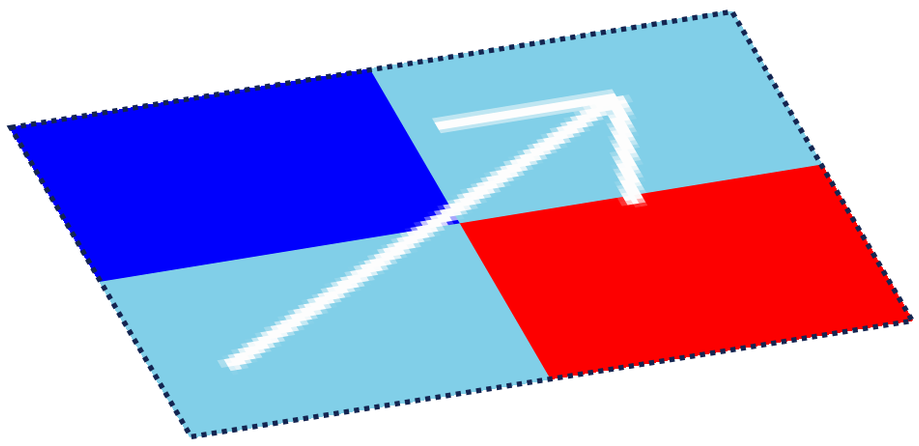
Expansion / compression / nonuniform scale



Signed dimensions also support horizontal / vertical mirroring.

PlgBlt

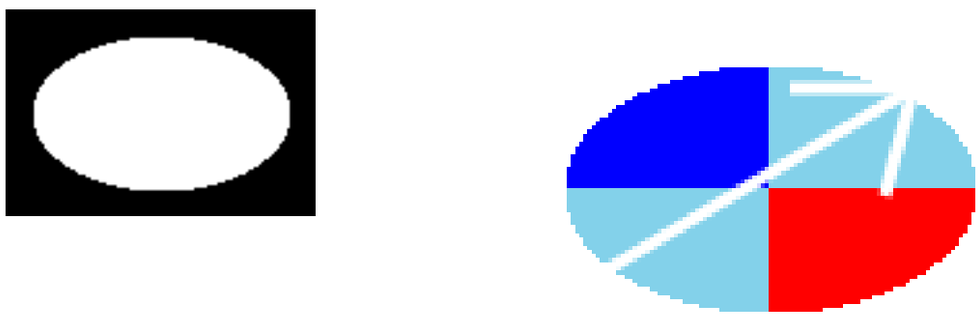
Three points define the affine destination parallelogram



Source origin + two basis vectors determine the warp.

MaskBlt

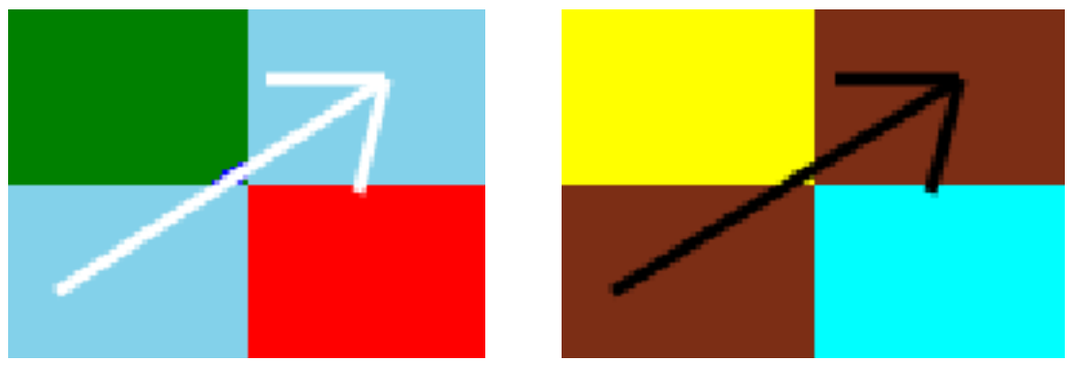
White mask copies source; black keeps destination



MaskBlt works on all targets, with clipping and transforms.

BrushCopy + SRCINVERT

Blue key becomes brush color; right sample uses XOR



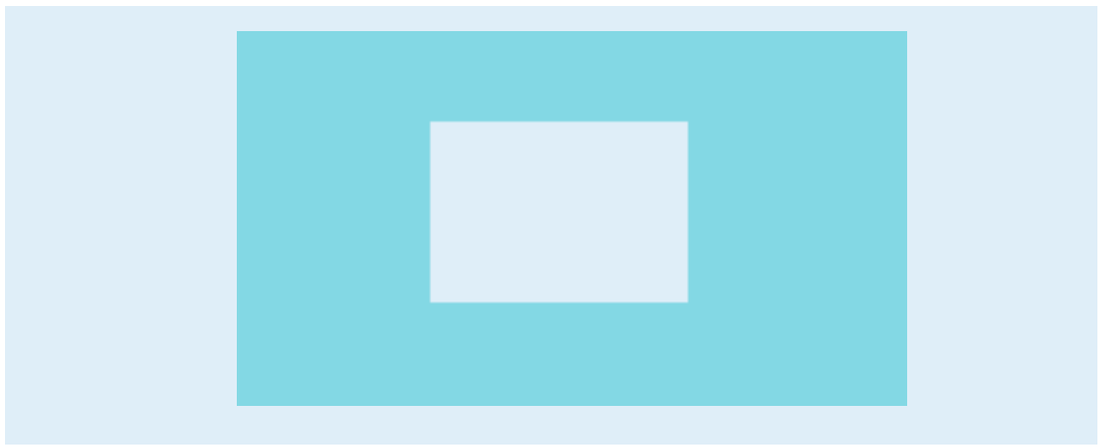
Color-key substitution and destination-dependent ROP are active.

Skia canvas / complete API atlas

06-clips-transforms / native results and compatibility operations

Set / intersect / exclude clip

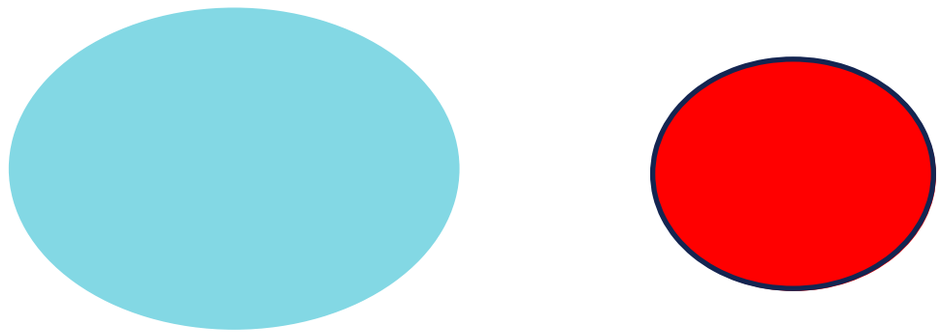
SetClipRect / IntersectClipRect / ExcludeClipRect



Automatic difference-clip fallback; bounds 260x145

Region clip + ResetClip

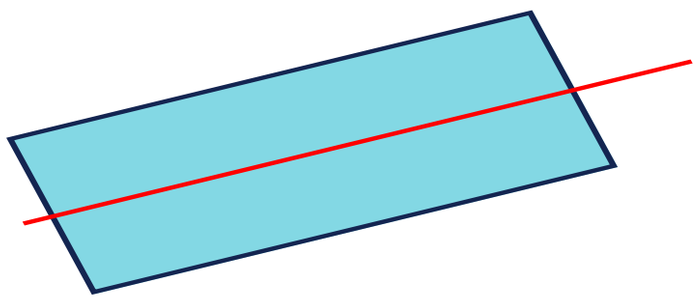
SetClipRegion(TRect) / reset / draw outside old clip



Arbitrary path/region clips; ResetClip removes the old shape.

Rotate / scale / translate

Transform order is explicit: local -> rotated -> device



All geometry shares the same world transform.

World matrix + inverse coordinates

Set/Get/ResetWorldTransform / LPToDP / DPToLP

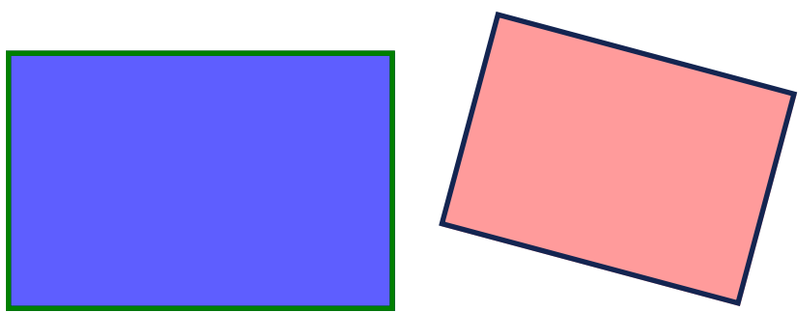


(50,30) -> (658,652) -> (50,30)

Shear M21 = 0,55; origin restored after Reset.

Nested SaveState / RestoreState

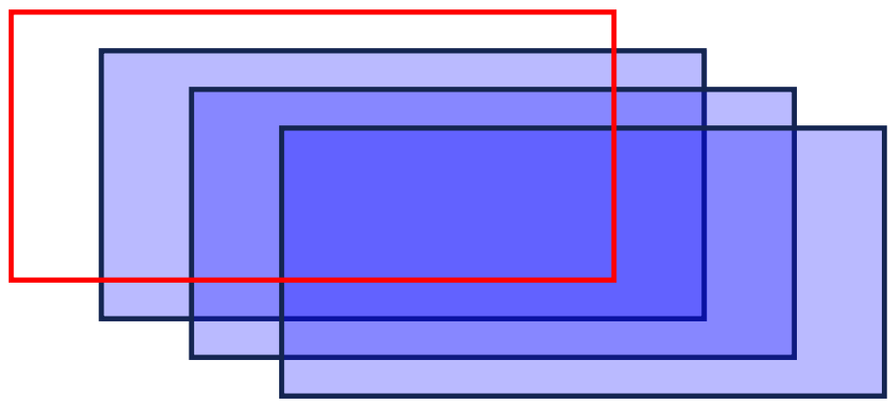
Style, matrix, pen position, text mode and clip restored



LIFO restore: origin (0,0), Brush.Alpha=255

Indexed restore / discard inner states

RestoreState(index) unwinds all younger snapshots



Red outline: original coordinates after indexed unwind.

Skia canvas / complete API atlas

07-text-fonts / native results and compatibility operations

Font rasterization options

FontEdging / FontHinting / FontSubpixel

Alias / None / subpixel off

AntiAlias / Normal / subpixel on

SubpixelAntiAlias / Full

Raster glyph settings; recording canvases retain vector outlines.

TextRect alignment + clipped overload

tfLeft / tfCenter / tfRight / explicit X,Y

TextRect alignment

TextRect alignment

TextRect alignment

Explicit coordinates

See page 9 for wrapping, ellipsis, tabs and prefix handling.

Font styles + alpha

Bold / italic / underline / strikeout / transparency

Font style / alpha

Font style / alpha

Font style / alpha

~~Font style / alpha~~

Decorations are drawn by the canvas, not by the font.

Metrics + tfCalcRect + assignment

TextExtent / TextWidth / TextHeight / AcquireFont

Measure AV

Extent 129x27; tfCalcRect 126x27

Width=129 Height=27 Ascent=-21,7

TFont.Assign reads VCL properties / real LOGFONT handles.

Custom typeface registration

LoadTypefaceFromFile / FromStream / register / unregister

Custom typeface: 012345

Fallback: 012345

Custom faces support shaped outlines and formatted text.

Opaque / transparent text backgrounds

Brush.Style bsSolid vs bsClear; clipped rendering

Opaque background

Transparent background

Rotated

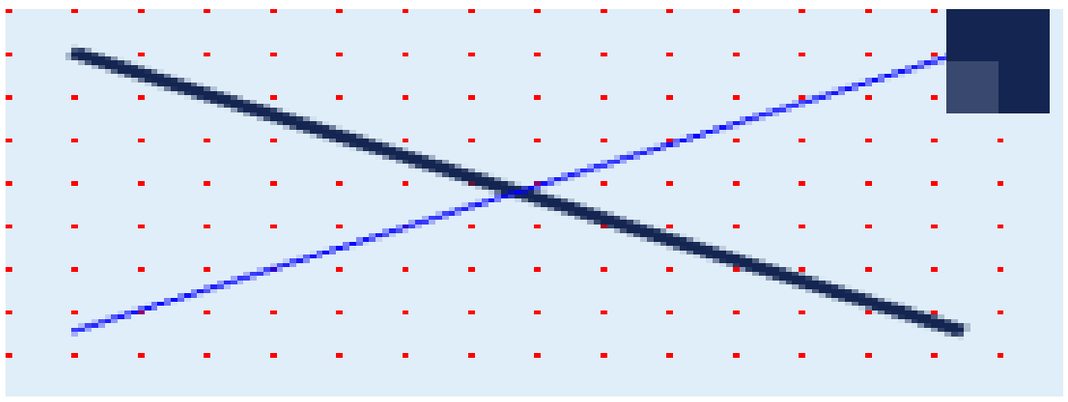
Brush paint follows the measured shaped text bounds.

Skia canvas / complete API atlas

08-raster-resources / native results and compatibility operations

Pixels / Flush / Snapshot

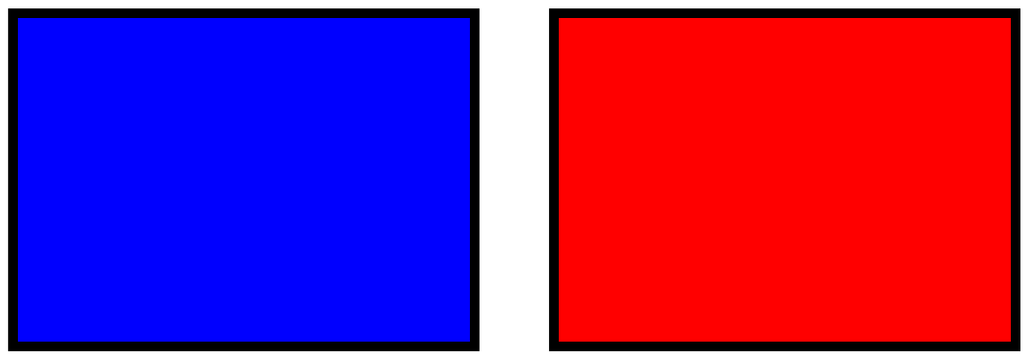
A bitmap canvas exposes pixel reads and writes



Pixels / Flush / Snapshot / ReadPixels / DrawPixmap.

FloodFill: surface and border

All targets; seed and boundary color select the filled area



fsSurface fills the seed color; fsBorder stops at Color.

Resource manager / cache lifecycle

Fill / stroke / font caches and explicit invalidation

Same fill key: True

After invalidation: True

Stroke / font acquired: True

OnChange events: 1; color round-trip=$FF

Cached interfaces remain valid after manager invalidation.

Glyph cache + outline metrics

GetLogFont / FontKey / GetEmSquare / ShapeText / Clear

EM=2048 / actual character height=32

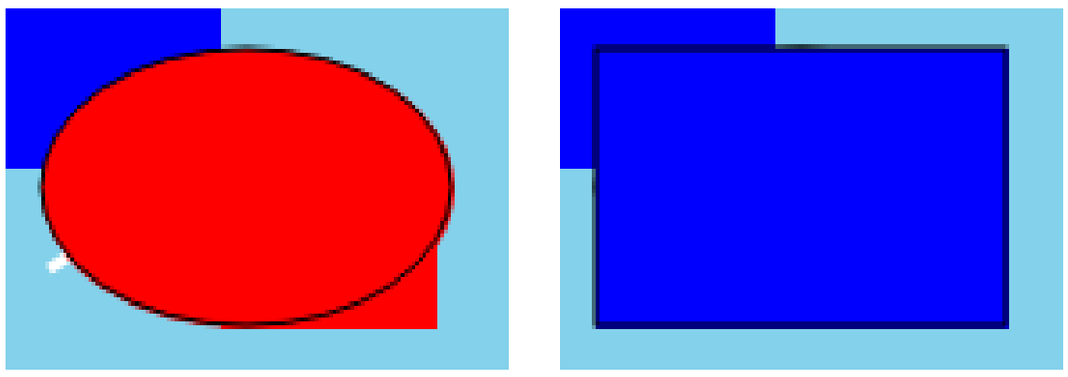
A: advance=1366,0; outline 1372,0x1466,0

LOGFONT key: Arial|-32|400|0|0|0|1|0

GDI paths use font design units and reusable font/DC resources.

Metafile / playback / serialization

FinishRecording / both Playback overloads / SerializeToStream



Serialized picture: 409 bytes; width=140, height=90

Parsed EMF + seeded surfaces

EMF records / glyph contours / direct Skia surfaces



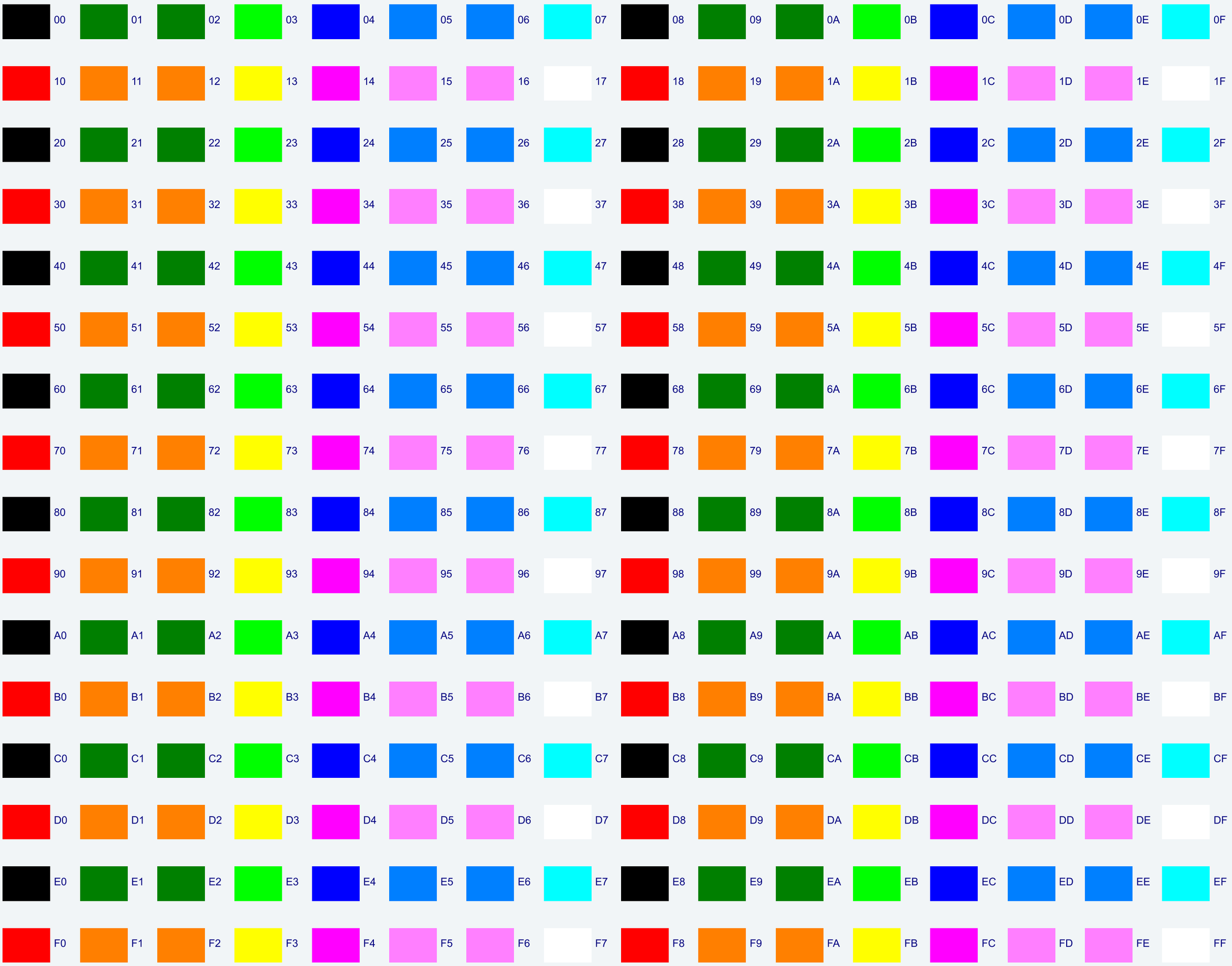
Draw / StretchDraw: EMF parser and recorded glyph contours

Destination reads use a mirrored surface; output stays portable.

Skia canvas / complete API atlas

09-rop-formatting / native results and compatibility operations

All 256 ROP3 truth tables (P = red, S = blue, D = green)



Wrapped text with kerning:
AVATAR office. Tabs: 123

A long fil...\shaped-outline.svg

Skia canvas / complete API atlas

10-unicode-fonts / native results and compatibility operations

Script-aware font matching

MatchTypefaceForText verifies every visible scalar

en -> Consolas

Latin AVATAR

ar -> Tahoma

العربية

zh-Hans -> Microsoft YaHei

中文字体

ja -> Yu Gothic

日本語かな

TTC / CJK face mapping

Read collection directory; select by glyph coverage

simsun.ttc / 2 indexed faces

Face 0: SimSun

Face 1: NSimSun

中文 / 漢字

Coverage-selected face index: 0

UTF-8 -> Unicode scalars

Offsets are bytes; supplementary scalars stay intact

offset / bytes / code point / script

1 / 1 / U+0041 / Latin

2 / 2 / U+00E9 / Latin

4 / 3 / U+4E2D / Han

7 / 4 / U+1F600 / Common

ZWNJ is default-ignorable: True

Mixed RTL paragraph

Logical Unicode input -> OpenType visual glyph order

ABC 123 / שלום / سلام

Automatic base level: 0

Use tfRtlReading for an explicit RTL paragraph base direction.

Arabic + Persian contextual shaping

Internal shaper: Uniscribe OpenType API

سلام لا

7 UTF-16 units -> 5 OpenType glyphs

فارسی / می‌روم

14 UTF-16 units -> 14 OpenType glyphs

OpenType pipeline + Unicode properties

ScriptItemizeOpenType / ShapeOpenType / PlaceOpenType

AVATAR office

Last run used OpenType: True

Adlam U+1E900: Adlam

RTL: True

Scalar round-trip: 1E900

Skia canvas / complete API atlas

11-unicode-rtl / native results and compatibility operations

Unicode 17.0: all 37 scripts containing assigned R/AL letters

The installed font set determines glyph availability. Missing coverage is reported explicitly.

Adlam / U+1E900
Ebrima



Arabic / U+0620
Tahoma



Avestan / U+10B00
Sans Serif Collection



Chorasmian / U+10FB0
Sans Serif Collection

Cypriot / U+10800
Segoe UI Historic



Elymaic / U+10FE0
Sans Serif Collection



Garay / U+10D4A
Ebrima



Hanifi_Rohingya / U+10D00
Sans Serif Collection



Hatran / U+108E0
Sans Serif Collection



Hebrew / U+05D0
Segoe UI



Imperial_Aramaic / U+10840
Segoe UI Historic



Inscriptional_Pahlavi / U+10B60
Segoe UI Historic



Inscriptional_Parthian / U+10B40
Segoe UI Historic



Kharoshthi / U+10A00
Segoe UI Historic



Lydian / U+10920
Segoe UI Historic



Mandaic / U+0840
Sans Serif Collection



Manichaean / U+10AC0
Sans Serif Collection



Mende_Kikakui / U+1E800
Sans Serif Collection



Meroitic_Cursive / U+109A0
Segoe UI Historic



Meroitic_Hieroglyphs / U+10980
Sans Serif Collection



Nabataean / U+10880
Sans Serif Collection



Nko / U+07CA
Ebrima

Old_Hungarian / U+10C80
Sans Serif Collection



Old_North_Arabian / U+10A80
Sans Serif Collection



Old_Sogdian / U+10F00
Sans Serif Collection



Old_South_Arabian / U+10A60
Segoe UI Historic



Old_Turkic / U+10C00
Segoe UI Historic



Old_Uyghur / U+10F70
Sans Serif Collection

Palmyrene / U+10860
Sans Serif Collection



Phoenician / U+10900
Segoe UI Historic



Psalter_Pahlavi / U+10B80
Sans Serif Collection

Samaritan / U+0800
Sans Serif Collection

Sidetic / U+10940
No installed font covers this scalar

Sogdian / U+10F30
Sans Serif Collection



Syriac / U+0710
Segoe UI Historic



Thaana / U+0780
MV Boli



Yezidi / U+10E80
Sans Serif Collection

Skia canvas / complete API atlas

12-text-on-path / native results and compatibility operations

Fit text to a cubic curve

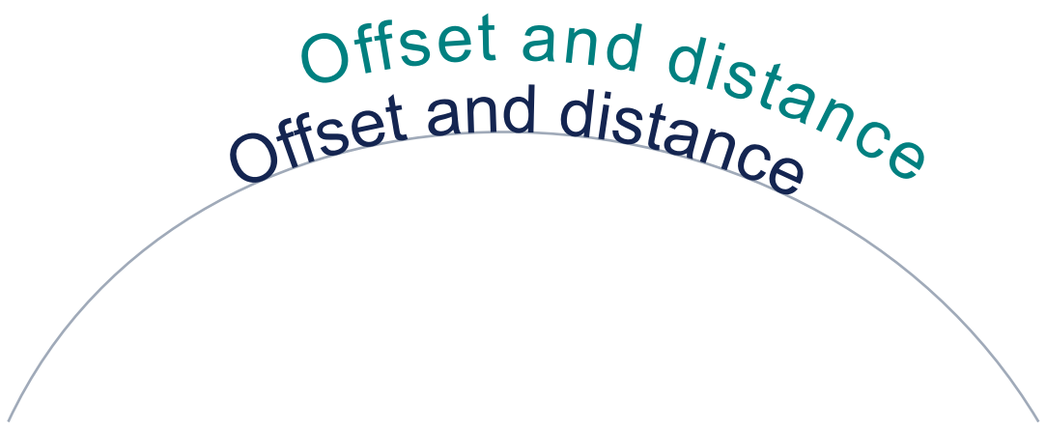
Tangent orientation / centered shaped text



TextOnPath / vector outlines in BMP, PDF and SVG

Distance and offset

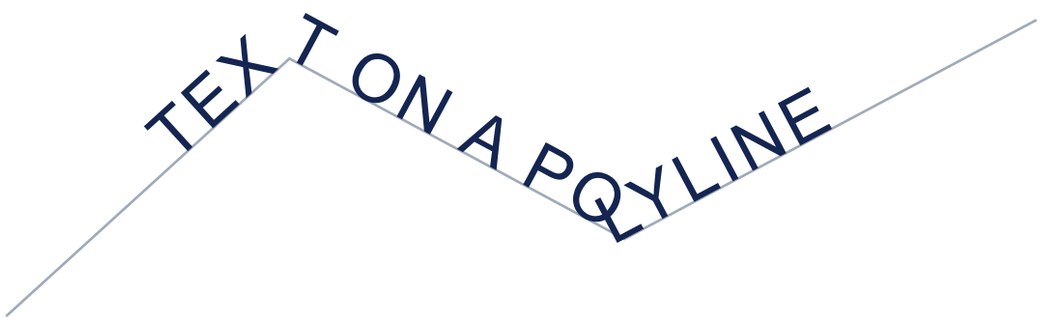
Same path / +28 normal / +35 along path



TextOnPath / vector outlines in BMP, PDF and SVG

Polyline

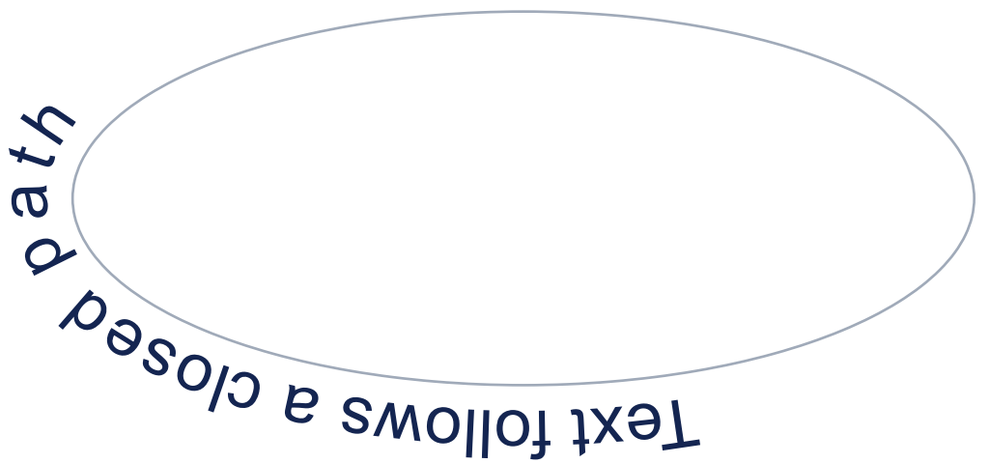
Repeated vertices / sharp bends / letter spacing



TextOnPath / vector outlines in BMP, PDF and SVG

Closed path

Wrap around an ellipse / negative offset



TextOnPath / vector outlines in BMP, PDF and SVG

Reverse and mirror

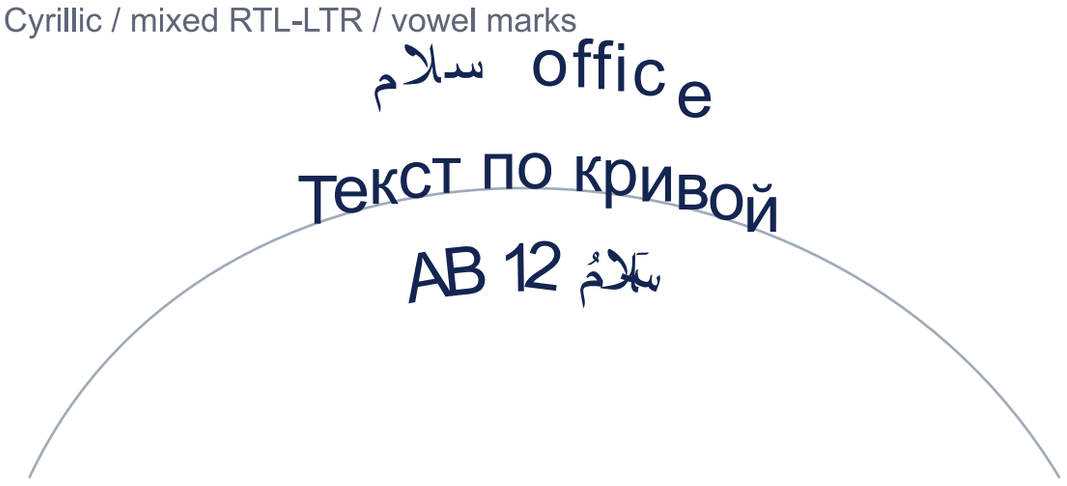
Reverse direction + vertical mirror / outside



TextOnPath / vector outlines in BMP, PDF and SVG

Unicode and orientation

Cyrillic / mixed RTL-LTR / vowel marks



TextOnPath / vector outlines in BMP, PDF and SVG